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SHORT NOTE

Inferring fast ecotypic divergence in a protected marine area: comparing Q_{ST} and F_{ST} patterns in *Littorina saxatilis* subpopulations from Cíes Islands in Spain

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Abstract

The intertidal snail *Littorina saxatilis* is well known for its remarkable variety of locally adapted populations. A unique *L. saxatilis* habitat is represented by the sheltered rocky environment found in the natural coastal lagoon of the Cíes Islands (NW Spain). The present-day characteristics of this habitat are relatively recent (~ 70–150 years), some of these have been human induced. In this study, we indirectly tested local adaptation resulting in ecotypic variation in this habitat by estimating the additive genetic differentiation between populations (Q_{ST}) for morphological shell traits in families of embryos. We compared the Q_{st} and F_{st} (AFLP loci) between populations of the previously described Crab ecotype (outside the lagoon) and the new Cíes lagoon ecotype (inside) as a proxy for adaptation. For several traits (RW_1 and RW_2; associated with a change in size of shell aperture) the Q_{ST} was higher than corresponding F_{ST}, which indeed suggest that natural selection could be the force behind the apparent phenotypic divergence observed between the Crab and the Cíes lagoon ecotype. The putative process of local adaptation occurring in the Cíes lagoon ecotype (favoring a reduction of shell aperture) might be considered a case of fast phenotypic evolution driven by anthropogenic effects and with important conservation implications. Further genetic and ecological studies must be carried out to advance our knowledge in such an interesting system.

Introduction

The rough periwinkle, *Littorina saxatilis* (Olivi 1792), is a dioecious intertidal gastropod found in a wide range of habitats across the North Atlantic (Reid 1996; Rolán-Alvarez et al. 2015), including exposed and sheltered rocky shores, cliffs, boulder beaches, coastal lagoons and brackish habitats. The species is ovoviviparous, having the females a brood pouch where from dozen to hundreds of embryos are developed until they are released as crawl-away juveniles that resemble miniature adults (Reid 1996). Low lifetime dispersal results in ecotype formation that are well adapted to divergent environments (Behrens Yamada 1989; Rolán-Alvarez et al. 2015), making classical taxonomy using morphological characters difficult (Johannesson and Johannesson 1990; Reid 1996; Gosling et al. 1998).

The use of molecular markers has shown that *L. saxatilis* populations can occasionally diverge genetically, even if they are separated only by a few meters of rocky shore (e.g., Galindo et al. 2013, Kess et al. 2018). In a few well-known cases, a strong ecological divergence between two subpopulations (e.g., ecotypes) caused a partial genetic barrier between them as a result of the divergent selection. In these cases, specimens from one of the ecotypes can hardly survive and mate in the habitat of the other ecotype because they are adapted to their specific habitat (Rolán-Alvarez 2007). However, in other cases, subpopulations showing particular phenotypes have been observed and described although the evolutionary mechanism responsible is still unknown (e.g., Johannesson and Johannesson 1990; Gosling et al. 1998). This evolutionary mechanism could be inferred either by studying the genetic basis of the trait (Stapley et al. 2010) or through manipulative experiments in the field, although both kinds of studies are difficult to conduct and

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have only been accomplished in a few referenced populations and traits (e.g., Boulding et al. 2017; Westram et al. 2018).

Common garden experiments, which imply breeding multiple families in controlled conditions, are another classical approach to partition the genetic and environmental contributions to quantitative traits (Hill 2010; de Villemereuil et al. 2016). This strategy allows the comparison between population genetic (additive) differentiation for quantitative traits (Q_{ST}) and between population genetic divergence for neutral molecular markers (F_{ST}). Such comparison has been used as an indirect test for the relative role of genetic drift and natural selection driving quantitative trait differentiation (Merilä and Crnokrak 2001; Leinonen et al. 2013); for example, both statistics should be similar ($Q_{ST} = F_{ST}$) under the assumption that both the quantitative trait and the molecular markers are preferentially affected by genetic drift (neutral), while the two estimates should differ ($Q_{ST} > F_{ST}$) when the quantitative trait is affected by divergent natural selection. This approach, however, is not feasible in those organisms that do not easily breed in laboratory conditions. In the case of *L. saxatilis* and other ovoviviparous organisms, there is an alternative, which involves the study of traits from shelled embryos across different families (i.e., females). The Q_{ST} estimated with this approach is a good proxy of the Q_{ST} of the same trait in adults. This design has been used several times in *L. saxatilis* (Newkirk and Doyle 1975; Conde-Padín et al. 2007; Galindo et al. 2019). Those studies have systematically shown higher Q_{ST} for different shell traits compared to the genetic divergence at neutral molecular markers between ecotypes (Conde-Padín et al. 2007; Galindo et al. 2019). Additionally, this extreme polymorphism of *L. saxatilis* is found replicated in parallel across different locations with similar habitats, which is an important step towards showing adaptive variation (Butlin et al. 2014; Westram et al. 2016; Galindo et al. 2019).

On the other hand, certain habitats could have rather unique characteristics and, therefore, replication in nature could be difficult. Nevertheless, if specific phenotypes are associated with these kinds of habitats, they may represent interesting populations from the point of view of conservation. Additionally, they could also add important knowledge to the understanding of the processes of local adaptation in this species and in general, as well as to the evolutionary mechanisms that promote and maintain genetic variation. An interesting example of these rare habitats is the case of the coastal lagoon (3600 cal BP; Costas et al. 2009) found in the Cíes Islands (Ría of Vigo, NW Spain; Fig. 1). This archipelago belongs to the Atlantic Islands of Galicia Maritime-Terrestrial National Park and it is considered a protected natural reserve (BOE 2002). The barrier–lagoon complex is confined by the 1-km-long beach–dune system

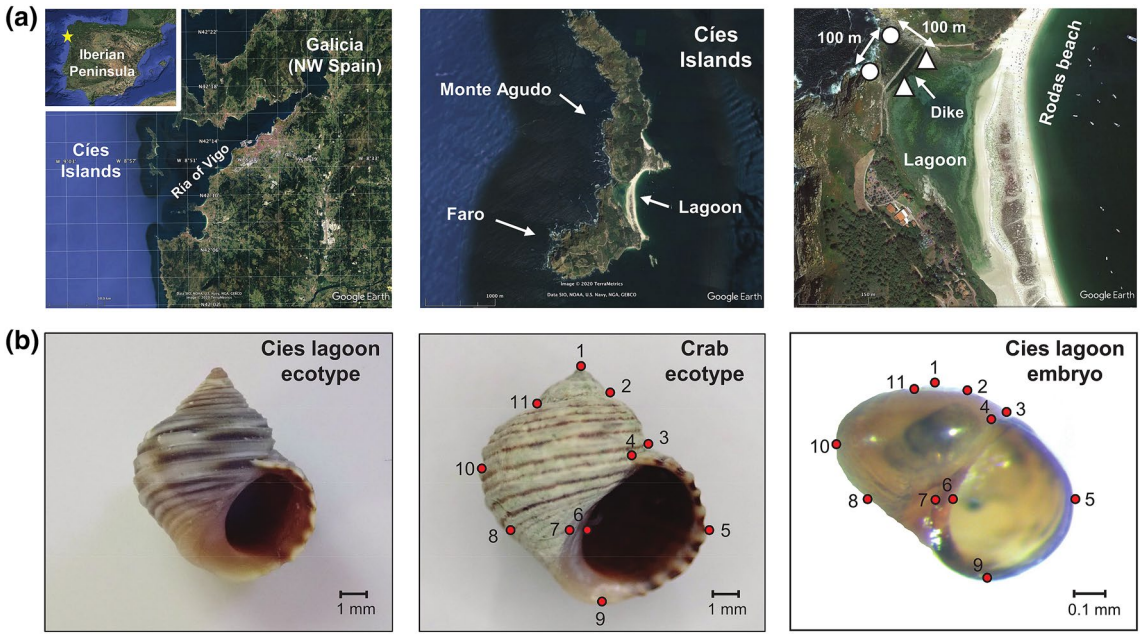


Fig. 1 **a** Geographical location of the Cíes archipelago within NW Spain, the coastal lagoon formed between the islands of Faro and Monte Agudo, with the West barrier represented by the nineteenth century dike, and with the Rodas dune–beach system on the east side of the lagoon. Triangles and circles represent the location of the sampled subpopulations of the *L. saxatilis* Cíes lagoon ecotype (inside the lagoon, triangle) and the Crab ecotype (outside, circle), respectively. **b** A typical specimen of the Cíes lagoon ecotype, the Crab ecotype and a *L. saxatilis* shelled embryo. The landmarks (LMs) used in the geometric morphometric analysis (LM1–LM11) are shown both in the Crab ecotype and *L. saxatilis* shelled embryo

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of Rodas that joins the eastern side of the Monte Agudo and Faro islands. The west side of the lagoon is confined by a 2.4 m elevation of the bedrock above local mean sea level (Muñoz Sobrino et al. 2016). Furthermore, the lagoon has suffered important changes since the late nineteenth century with the construction of a dike that joins the bedrock of both islands in the west, creating a more sheltered environment during storms and a more restricted exchange of water with the ocean. In addition, the opening of ephemeral inlets in the east has been changing since the 1950s due to the reshaping of the dune–beach system. Until that time a southern ephemeral inlet, that worked as a drainage channel, has slowly disappeared and since the enlargement of the Rodas dock in 2010 a sporadic smaller inlet occurs in the northeast extreme of the lagoon during spring tides and storms (Costas et al. 2002; Muñoz Sobrino et al. 2016; Gómez-Pazo et al. 2019). The *L. saxatilis* habitat within the lagoon is restricted to a highly sheltered bolder field close to the dike, with delayed and less pronounced tides due to the connection to the outside through a small opening below the dike. On the western part of the islands, outside the lagoon, the habitat is highly exposed and Crab and Wave ecotypes (see Butlin et al. 2014) are found. Therefore, < 100 m from the bolder field inside the lagoon and on the other side of the dike it is possible to find populations of the Crab ecotype and these are, apparently, phenotypically different from the lagoon morph.

The aim of the study is to compare patterns of genetic differentiation to detect traits driven by divergent natural selection (Storz 2002). We studied two replicates of each population, the Crab ecotype and the lagoon morph from Cíes Islands, and compared levels of population genetic differentiation for different shell quantitative traits (Q_{ST}) and AFLP molecular markers (F_{ST}), to infer the evolutionary mechanism shaping the difference between these populations.

Materials and methods

Study site and experimental design

Samples of *L. saxatilis* were collected at Cíes Islands (42°13′23.7"N, 8°54′17.1"W; NW Spain, Fig. 1a), located at the entrance of the Ría (estuary) of Vigo, in October 2017 and March 2018. We sampled two different habitats: the first habitat is a sheltered bolder field inside the Cíes lagoon, close to the western barrier of the lagoon represented by a dike and also a natural elevation of the bedrock of the islands, compared to the sea level. The shell traits of this subpopulation (lagoon morph; Cíes lagoon ecotype, hereafter) have not been previously characterized. The second habitat is found outside of the lagoon on the exposed (West) side of the islands. In this habitat we collected the Crab ecotype in the upper-shore barnacle zone (*Chthamalus stellatus*), separated approximately 100 m from the bolder field in the lagoon. For each habitat, two replicate subpopulations were sampled (four in total, Fig. 1a). The distance between replicates within ecotype is similar to the distance between ecotypes. Approximately 150 snails were collected for each subpopulation (in about 2–4 m^2 for Crab and 40 m^2 for Cíes lagoon ecotypes, correspondingly), these were brought alive to the laboratory to process them.

Sample processing

The shell of a random set of individuals from each subpopulation was photographed using a Leica digital ICA video camera mounted on a Leica MZ12 stereoscopic microscope (Leica Microsystems), as described in Conde-Padín et al. (2007). Then they were dissected and sexed under the microscope. The head–foot tissue was stored at -20 °C until further genetic analysis. The brood pouch of the females was dissected and the shelled embryos were stored in ethanol at 4 °C, independently for each mother. The shell imaging and dissections were carried out until 15 brooding females were obtained for each subpopulation. Following (Galindo et al. 2019), from each female, three random embryos from the latest stages of development (dark shell) were selected for shell imaging, applying the same methodology used for adults.

Morphometric analysis

The morphometric analysis was performed with the TPS software suite (https://life.bio.sunysb.edu/morph/; Rohlf 2015) using 60 females (15 per subpopulation) and 180 embryos (three per female; 15 families per subpopulation). Following (Conde-Padín et al. 2009) 11 landmarks (LMs) were positioned on the shell images using TpsUtil v.1.44 and TpsDig v.1.40 (Fig. 1b). The analysis was carried out independently for shelled embryos and adult females. Then TpsRelw v.1.46 was used to estimate the centroid size (CS, the square root of the sum of squared distances of the LMs to their centroid; Bookstein 1991) and perform the relative warp analysis as described in (Galindo et al. 2019). Briefly, shell shape was estimated by aligning raw specimen coordinates using the Procrustes generalized orthogonal method (GLS; Rohlf and Slice 1990), which generates a reference configuration by minimizing the squared distances between homologous landmarks across individuals. Then the linear projections of deviations for each individual from the reference configuration in the multidimensional space of shape, the partial warp scores, are calculated. The principal components of the partial warp scores are finally obtained and are referred to as relative warps (RWs; see Zelditch et al. 2004). The first four relative warps (RW_1 to RW_4) explained more than 87% of the variation in shape in adults and around

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75% in embryos. Therefore, they are highly representative of the overall shell shape variation and further analyses only included these RWs and the CS. The biological interpretation of the RWs was done using the thin-plate spline representation of the studied samples (see "Results and discussion"), also obtained with TpsRelw v.1.46. The relative shell aperture area was estimated on the images captured from the thin-plate spline representation using the software KLONK Image Measurement 15.1.1 (Andersen 1993).

Genetic analysis

Genomic DNA was isolated from head–foot tissue of 30–40 individuals per subpopulation (including the 60 females analyzed for geometric morphometrics) using the E.Z.N.A® Mollusc DNA Kit (Omega Bio-Tek), following the manufacturer's instructions. DNA was quantified using BioDrop µLITE (Biodrop) and quality was scored on 2% agarose gels. AFLP (Amplified Fragment Length Polymorphism) analysis was performed using a modified version of the protocol described by Galindo et al. (2009). Adaptor, primer sequences and PCR cycling conditions are described in the same study. For each individual, a DNA aliquot of 50 ng was digested and ligated using 5 U of *Eco*RI and 3 U of *Mse*I (NEB, New England Biolabs), 5 pmol Eco adaptor, 50 pmol Mse adaptor and 0.4 U of T4 DNA ligase (NEB) in 20 µL total volume of 1 × NEB buffer and 1 × ligation buffer (NEB) supplemented with 2.5 µg of BSA for 2 h at 37 °C. Preselective PCRs contained 4 µL of 1:10 digestion–ligation–dilution, 10 µL of NZYTaq II 2 × Colourless Master Mix (NZYTech) and 20 pmol Eco + A and Mse + C preselective primers. Then, selective PCRs were performed with 4 µL of 1:10 preselective PCR dilution, 10 µL of NZYTaq II 2 × Colourless Master Mix (NZYTech) and 8.3 pmol of Eco + ACT and Mse + CAC selective primers. All PCRs were performed in a final volume of 20 µL. The selective PCR was electrophoresed along with GeneScan 500 ROX on an ABI3130 Genetic Analyzer (Thermofisher) at CACTI (Centro de Apoyo Científico y Tecnológico a la Investigación) genomic facilities at the University of Vigo. AFLP genotyping was performed using the software GeneMapper v.3.7 (Thermofisher). The program AFLP-SURV v.1.0 (Vekemans et al. 2002) was used to estimate pairwise F_{ST} values following (Lynch and Milligan 1994) and their significance by 10,000 random permutations of individuals among populations.

Estimates of Q_{ST} and P_{ST}

Phenotypic and quantitative genetic differentiation (P_{ST} and Q_{ST}) were estimated on the morphological variables (CS and $RW_{1\text{-}4}$) from adult females and shelled embryos, correspondingly, and assuming a half-sib relationship between the embryos within each female. (Conde-Padín et al. 2007) assumed full-sibs in their analyses. However, (Galindo et al. 2019) and this study have assumed half-sibs because (Panova et al. 2010) determined that the progeny of a single female could be sired by 15–23 males.

A two-way ANOVA was used on the morphological (dependent) variables (CS and $RW_{1\text{-}4}$), together with the fixed factor *Subpopulation* (between or within ecotype depending on the comparison) and *Family*. The Q_{ST} was estimated using the former variance components applying the following the equation:

$$Q_{ST} = \frac{V_{bpe}}{V_{bpe} + \left(\frac{2V_{wpe}}{(1+2\rho h^2+\rho^2 h^2)}\right)} \quad (1)$$

where h^2 is the heritability of the trait averaged between populations (Galindo et al. 2019; Eq. 1), and V_{wpe} and V_{bpe} are the variance components within and between populations from a two-way ANOVA. The ρ is the observed Pearson correlation coefficient of the trait in mating pairs in the population (CS = 0.5; $RW_{1\text{-}4}$ = 0; see Galindo et al. 2019). Note that $V_{wpe} = 4V_{bf}$, i.e., the variance component between families of embryos (factor family). There is, however, a non-negligible probability of occasionally including two full-sib embryos ($P = 0.1244$, assuming 15 fathers on average, see Panova et al. 2010) within each family, instead of three half-sibs. In this case, our estimate of the variance between families of embryos (V_{bf}) would be biased upwards and, therefore, the Q_{ST} would be biased downwards (because V_{wpe} is in the denominator; see further discussion in Galindo et al. 2019). This scenario would reduce our chance to detect significant Q_{ST}, as well as the chances of Q_{ST} being higher than F_{ST}, and so we consider our analysis to be conservative.

Similarly, the P_{ST} was estimated as

$$P_{ST} = \frac{V_{bpa}}{V_{bpa} + \left(\frac{2V_{wpa}}{(1+2\rho h^2+\rho^2 h^2)}\right)} \quad (2)$$

where h^2 is the heritability (Galindo et al. 2019; Eq. 1), and V_{wpa} and V_{bpa} are the phenotypic variance components within and between populations of the adult females analyzed.

Two replicates of Crab *vs*. Cíes lagoon ecotypes were used to study quantitative genetic (Q_{ST}) and phenotypic (P_{ST}) differentiation. Significance P_{ST} and Q_{ST} were obtained by corresponding *F* test from ANOVA or *t* test on averages between groups (Sokal and Rohlf 1995). The statistical analyses were carried out in SPSS ver. 24 (IBM Corp) using the GLM (general linear model) function and type III sum of squares, being especially robust to deviations from ANOVA assumptions.

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Results and discussion

The comparison of *L. saxatilis* subpopulations from inside and outside the lagoon at Cíes Islands (NW Spain) represents an interesting model system to study whether the existing morphological differences between them could be potentially caused by recent forces of natural selection, resulting in fast ecotypic divergence.

A total of 136 individuals were analyzed for 249 AFLP loci. The analysis of AFLPs, besides being a dominant marker and being subjected to certain level of homoplasy, allows the efficient estimation of the level of genetic differentiation (F_{ST}) between subpopulations (Caballero et al. 2008; Table 1). The level of differentiation between habitats (Crab *vs.* Cíes lagoon ecotypes) was the same than that found between replicates within habitats, and although these levels of differentiation were low, they were significant ($P < 0.05$, after sequential Bonferroni corrections) in all comparisons. These F_{ST} values were also similar to those obtained for AFLPs in previous *L. saxatilis* studies when subpopulations shared habitat and were separated by less than 100 m (Piñeira et al. 2008; Galindo et al. 2009, 2013). Interestingly, this suggests that the apparent environmental differences between habitats do not represent an important barrier to gene flow between the ecotypes such as the one observed between Crab and Wave ecotypes (Rolán-Alvarez et al. 2004; Galindo et al. 2009; Butlin et al. 2014; Kess et al. 2018). The low genetic (molecular) differentiation between these ecotypes, the absence of a long-term physical barrier (the dike was built around 150 years ago) and the fact that the current habitat inside the Cíes lagoon is also new due to the recent changes in the dune–beach system, makes the existence of reproductive barriers between these ecotypes improbable. In fact, even the low genetic differentiation between the Crab and Cíes lagoon ecotype it is unlikely to be the result of incomplete lineage sorting (ILS) as it has been claimed between sibling species of *Littorina* (Wilding et al. 2000). For example, the Crab and Wave ecotypes, which show a relatively old ecological barrier, are not characterized by ILS (Rolán-Alvarez et al. 2004; Quesada et al. 2007; Butlin et al. 2014).

The pattern of genetic variation (Q_{ST}) within and between ecotypes (Crab *vs.* Cíes lagoon) is shown in Table 1, and for certain traits, it was considerably higher than the neutral molecular differentiation (F_{ST}). For example, Q_{ST} differentiation between ecotypes (Crab *vs.* Cíes lagoon) was high on average (0.18; range 0.04–0.46). The trait RW_1 showed the greatest level of differentiation. Actually, this was much higher than the Q_{ST} values between replicated subpopulations within ecotype (average = 0.04; range 0.03–0.05) or even F_{ST} (AFLPs) between ecotypes (Table 1). Then the fact that the genetic differentiation within ecotype for morphological traits (Q_{ST} *within*) is of similar magnitude to the one for neutral molecular markers (F_{ST} *within* and *between*), but Q_{ST} between putative ecological morphs (ecotypes) is considerable higher than the other two (Fig. 2), suggests that divergent natural selection is acting on the additive genetic variation in different directions in each habitat (sheltered inside the lagoon *vs.* exposed outside).

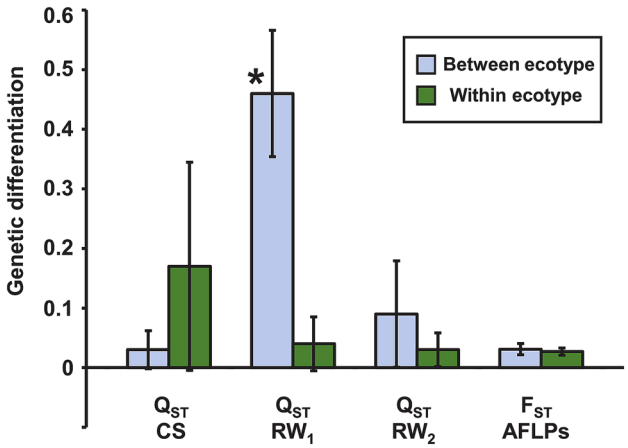


Fig. 2 Averaged values of genetic differentiation between and within ecotypes for morphological traits (CS, RW1 and RW2) and molecular markers (AFLPs) with standard error bars. * $P \leq 0.05$

Table 1 Patterns of within and between ecotype (Crab *vs.* Cíes lagoon) genetic differentiation for morphological traits (Q_{ST}; CS and RW_1 to RW_4, see "Materials and methods"; mean ± SD across subpopulations) and molecular markers (F_{ST}; AFLPs) and of phenotypic differentiation in adults (P_{ST})

Trait	Genetic differentiation (Q_{ST}, F_{ST})		Phenotypic differentiation (P_{ST})	
	Between	Within	Between	Within
CS	$0.03^1 \pm 0.045$	$0.17^1 \pm 0.247$	0.01 ± 0.004	$0.13^2 \pm 0.015$
RW_1	$0.46^2 \pm 0.150$	$0.04^1 \pm 0.064$	$0.09^2 \pm 0.053$	$0.09^2 \pm 0.049$
RW_2	$0.09^1 \pm 0.126$	$0.03^1 \pm 0.040$	$0.67^2 \pm 0.041$	$0.66^2 \pm 0.136$
RW_3	0.04 ± 0.050	0.03 ± 0.044	$0.06^1 \pm 0.057$	0.03 ± 0.044
RW_4	$0.14^1 \pm 0.195$	$0.03^1 \pm 0.059$	$0.05^1 \pm 0.069$	$0.14^1 \pm 0.176$
AFLPs	$0.031^4 \pm 0.019$	$0.027^2 \pm 0.012$		

[1–4]Number of significant ($P \leq 0.05$) estimates per comparison (four cases studied for F_{ST} and two for Q_{ST} and P_{ST})

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The pattern of differentiation in morphological traits, however, differs considerably in adults (P_{ST}; Table 1): first, the morphological trait with the greatest differentiation between ecotypes in adults was RW_2 instead of RW_1; and second, the differentiation showed similar levels of variation between and within ecotypes. Nevertheless, we should not compare Q_{ST} and P_{ST} directly because the RWs could represent different morphological traits as they have been studied in different life stages. The biological interpretation of the deformations of the RW_1 and RW_2 differed at both stages (Fig. 3): the differences between adult ecotypes affected mostly to shell spire and aperture; the Cíes lagoon ecotype commonly had a taller spire with a smaller aperture compared to the Crab ecotype. However, for shelled embryos the differences seem to preferentially affect the shell aperture and the area where the soft parts are accommodated within the shell (Fig. 3). The fact that different relative warps capture different shape characteristics in different developmental stages should be expected because the measurements were analyzed separately for adults and embryos and they are rather different in shape (Zelditch et al. 2004). Nevertheless, when both variables are simultaneously compared between embryos and adults some general change in shape is observed, affecting preferentially to the shell aperture (Fig. 3).

On the other hand, the pattern for RW_2 showing high phenotypic differentiation within ecotype (including high standard deviation) needs further clarification. A detailed examination of the pattern shows that it is caused by the Crab ecotype, which is something unusual and not observed in previous studies (e.g., Galindo et al. 2019). This might suggest that one of the collected Crab ecotype subpopulations could be characterized by a heterogeneous micro-habitat, and that this could be causing phenotypic plasticity to affect some of these morphological traits in adults. Nevertheless, we have to be cautious regarding this possibility, because the relationship of these traits between adults and embryos is still unknown and consequently we cannot infer the proportion of shell variation in adults explained by phenotypic plasticity.

In summary, our results show that shelled embryos from both ecotypes differ for quantitative genetic variation, and this cannot be caused by phenotypic plasticity, unless our Q_{ST} values would be biased by any environmental condition. Our Q_{ST} estimates were not produced from a common garden experiment in the laboratory, so they could be potentially environmentally biased into the between populations component of variation (the numerator from Eq. 1). However, this bias is highly unlikely in this study as the Q_{ST} has been estimated on embryos and these have not had time to experience the natural environmental conditions, as they

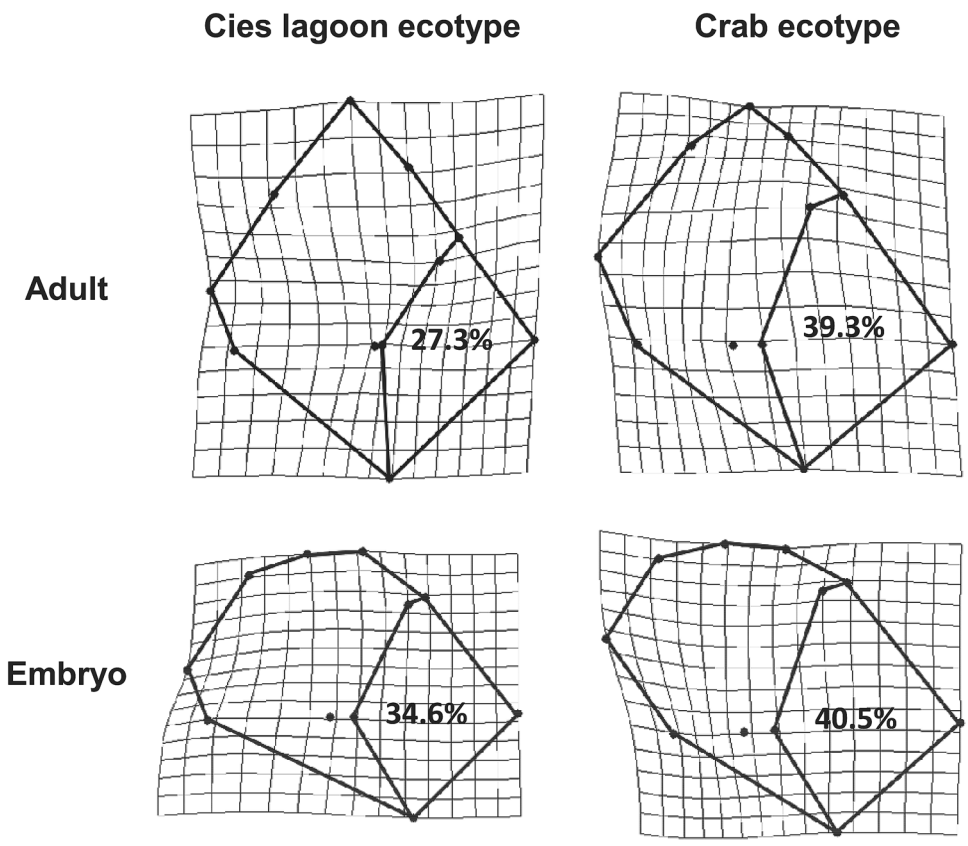


Fig. 3 Thin-plate spline representation of the average ± SD RW_1 and RW_2 within ecotype (Crab *vs*. Cíes lagoon) for embryos and adults. The landmarks in the deformation grid were manually joined by lines to give a better illustration of the shape of the shells of the two ecotypes. The values represent the relative shell aperture area (percentage of the total shell area)

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are growing inside their mother and, therefore, not expected to differ between ecotypes or subpopulations (Conde-Padín et al. 2007; Galindo et al. 2019). Furthermore, our estimate would be most probably biased downwards because of the occasional incorporation of full-sib embryos in our data (see "Materials and methods"). Nevertheless, ideally these Q_{ST} estimates should be confirmed by common garden experiments rearing families in the laboratory, if possible, or at least using transplant experiments to confirm the secondary role of the environmental condition, although these approaches are difficult in *Littorina* species. These results, combined with the lack of genetic structure for neutral molecular markers, represent an indication that natural selection might be playing a role in this divergence and that the subpopulations from the bolder field inside the Cíes lagoon could be considered another ecotype of *L. saxatilis* (Cíes lagoon ecotype). The Cíes lagoon ecotype has some resemblance with the Crab + ecotype from extremely sheltered habitats from the inner part of the estuaries in Galicia (NW Spain) (Tirado et al. 2016; Galindo et al. 2019), something expected given that the habitat inside the lagoon is also highly sheltered. Interestingly, in past studies another *Littorina* ecotype (*L. saxatilis tenebrosa* Montagu; Wilson et al. 1999; Small and Gosling 2000; but see Gosling et al. 1998) has been associated to coastal lagoons and brackish habitats. Nevertheless, the morphological description of this ecotype is quite different from the Cíes lagoon ecotype, as the former is smaller, more fragile (thin-shelled) and tall-spired.

Future genetic studies combining these two ecotypes would be needed to unravel their history. The Cíes lagoon ecotype is a particularly interesting case to study local adaptation, since the current characteristics of the Cíes lagoon are rather recent (~ 70–150 years old) (Gómez-Pazo et al. 2019). Therefore, these adaptations in *L. saxatilis* might be a few 100 s generations old, representing an example of contemporary evolution due to adaptation at a short timescale and most-likely human-induced. On the other hand, the Cíes archipelago, as part of a National Park, is a protected environment and the barrier–lagoon complex represents a specific area of the park where visitors are not allowed because of biodiversity conservation management program. For this reason the *L. saxatilis* intraspecific variation detected in this study, which could be due to the human-induced fragmentation of the habitat (e.g., dike construction), changes in the ocean water exchange of the lagoon (e.g., ephemeral inlets) or the unique characteristics and biodiversity of the lagoon, represent an important conservation unit that deserves special biodiversity attention. Further studies should focus on corroborating the role of natural selection by manipulative experiments (e.g., Boulding et al. 2017) as well as identifying the candidate genomic regions responsible for this putative local adaptation (e.g., Westram et al. 2018). The *L. saxatilis* Cíes lagoon ecotype represents an interesting system in which to study contemporary local adaptation in fragmented habitats and develop conservation management strategies.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest. The authors declare that all applicable international, national and/or institutional guidelines for sampling, care and experimental use of organisms for the study have been followed and all necessary approvals by the "Parque Nacional Marítimo Terrestre das Illas Atlánticas de Galicia" (Xunta de Galicia) have been obtained (107/RX317094).

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